

FRANCESC BENACH GARCIA

PROJECT MANAGER · ELECTRONICS ENGINEERING · EMBEDDED SYSTEMS · DIGITAL TRANSFORMATION & AI

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PROFESSIONAL PROFILE

Electronics engineer and project manager with experience in product development, hardware, C/C++ firmware and embedded systems. I have progressed from hands-on electronics design and programming to end-to-end project coordination, linking requirements, prototyping, manufacturing, validation and field support. I complement this background with applied training and projects in digital transformation, analytics, automation and artificial intelligence for SMEs.

CORE EXPERTISE

Technical project management and coordination · Hardware, schematic and PCB design · C/C++ firmware for ESP32, PIC and AVR · Embedded systems and IoT architecture
RF, NFC, sensors, audio and embedded UI · Digital transformation, CRM/ERP and traceability · Web applications, dashboards and automation · Generative AI, GPT agents and process improvement

PROFESSIONAL EXPERIENCE

DOIT Multisensorial | Project Manager · R&D Electronics Engineer

Aug 2023 – Present | Reus, Spain

- Progressed from electronics and project engineering into project-management responsibilities while retaining direct ownership of hardware and firmware.
- End-to-end development of multisensory products based on ESP32 and PIC: schematics, PCB, component selection, C/C++ firmware, prototyping, validation and manufacturing support.
- Development of New Gen and Legacy architectures with proprietary protocols, 868 MHz RF, NFC, displays, audio, sensors, batteries, local storage and physical user interfaces.
- Functional coordination of workshop and assembly technicians, defining build criteria, tests, diagnostics and issue-resolution actions without formal line-management authority.
- Cross-functional work with R&D, QA, management, suppliers and after-sales; preparation of BOMs, manuals, technical documentation and acceptance criteria.
- Drove and implemented the first phase of internal digital transformation through order traceability, Jira/Billage/Drive, dashboards, local applications and automation agents that remain in use.

PoweredCub | Electronics Engineer / Project Engineer

Apr 2021 – Jul 2023

- PCB and firmware design for electronic alarm and security systems used in industrial vehicles.
- Control-electronics development and validation, hardware selection, prototyping, functional testing and product-evolution support.

iDeeFe and Wits SL | R&D Electronics Engineer

Jun 2019 – Mar 2020 | Tarragona, Spain

- Developed a semi-automatic drum-filling prototype: PCB design with Altium Designer and KiCad, ATmega2560 firmware, actuators, sensors and motor control.
- Owned hardware selection, SMD assembly, validation and commissioning.

MAETEL | Electrical Drafter

Jun 2018 – Sep 2019 | Tarragona, Spain

- Drafted low-voltage electrical networks and photovoltaic connection sections using AutoCAD.

SELECTED PROJECTS

DOIT digital transformation. Applied project graded 10/10 and subsequently implemented internally: diagnosis, roadmap, COMXXXXXX traceability, Billage/Jira/Drive, KPIs, automation, backups, cybersecurity and applied AI.

DOIT CRM/Jira Dashboard. Standalone HTML, JavaScript and JSON dashboard consolidating orders, products and Jira data with operational search and filtering. Used internally on local systems.

DOIT Packing List App. Local application for allocating products to boxes, preventing duplicates, displaying package contents and handling partial lines; used in real operations.

Automation agents. Designed and validated GPT agents for packing lists, technical manuals, PCB-supplier emails, operational writing and document workflows, built with ChatGPT under my technical direction.

Lumina. Personal wellbeing web app with mood calendar, goals, medication, statistics and AI-assisted suggestions; React/TypeScript/Vite, local persistence and Gemini integration.

ESP32 Web Serial tools. Applications for NFC configuration/decoding with ESP32-S3 + PN532 and LED-matrix control through Web Serial, combined with C++ firmware and custom binary protocols.

Multisensory hotel-room control. Local web prototype with PIN/QR access to control lights, music, video, scenes, ventilation and alarms, plus room administration.

Tumburú. Local web application and digital-transformation proposal for a sports club, developed from an integral audit; non-commercial prototype.

EDUCATION & TRAINING

BEng in Industrial Electronics and Automation Engineering. Universitat Rovira i Virgili · 2013–2018

Additional Computer Engineering studies. Completed years 1 and 2 · Universitat Rovira i Virgili

Digital Transformation: AI and Big Data. GDPYMES Programme · Adquero Servicios Integrados SL · 150 hours · Mar–May 2026 · DOIT final project: 10/10

Introduction to AI. BIG School · ChatGPT, custom GPTs and Make · 4 hours · May 2025

Mastering RTOS: Hands-on FreeRTOS and STM32Fx with Debugging. Specialised real-time systems training

TECHNOLOGIES, LANGUAGES & AVAILABILITY

Firmware and programming: C, C++, Python, VHDL; Arduino, PlatformIO, MPLAB X, FreeRTOS

Hardware: Altium Designer, KiCad, OrCAD, Proteus, AutoCAD, MATLAB, PSIM, TIA Portal

Web and data: HTML, CSS, JavaScript, TypeScript, React, Vite, JSON, dashboards, Git/GitHub

Business and automation: Jira, Billage, Google Drive/Sheets, ChatGPT, GPTs, Make, n8n, CRM/ERP

Languages: Native Catalan and Spanish · Intermediate English · Basic French

Additional: Category B driving licence and own vehicle · Available for travel and hybrid/on-site work